MICROCONTROLLER

B1
STM32L052K6T6

VDD_1, PC14-OSC32_IN, PC15-OSC32_OUT, NRST, VDDA, PA1, PA2, PA14, PA13, PA12, PA11, PA10, PA9, PA8, VDD_2, VSS_2, BOOT0, PB7, PB6, PB5, PB4, PB3, PA15, PA3, PA4, PA5, PA6, PA7, PB0, PB1, VSS_1

C1 100nF, C2 100nF, C3 10uF 16V, C4 100nF, C5 100nF, C6 1uF 16V

SWCLK, SWDIO, USB_D_N, USB_D_P, SERIAL_RX, SERIAL_TX, SLEEP_WAKEUP, PWM_TIM2, BUTTON_IN_1, BOOT0, I2C_SCL, I2C_SDA, RST_SCREEN, ANT_DEMOD_3V3, ANT_SHD_3V3, ANT_LVLSHIFT_EN, BUTTON_IN_2, SPI_CS_EEPROM, SPI_CLK_EEPROM, SPI_MISO_EEPROM, SPI_MOSI_EEPROM, LED_R, LED_G

POWER PROTECTION

H3 — WM22663-ND (GND, VBATT, GND)

copied from joel's 461 board

PROCTECTION CIRCUITRY
- Reverse Polarity
(Overvoltage couldn't be achieved with the provided zener and was decided unecessary given the buck converter)

F1 5A, C17 100nF, SSM3J328R,LF (Q2), C19 100µF/25V, C20 100nF, C18 470uF/16V, R14 0R

VBATUNFUSED, VBATFUSED, VBATT

SCREEN HEADER

SCREEN — JA_PIN_Header 8

R10 10K, R11 10K, C15 1uF 16V, C16 1uF 16V

RST_SCREEN, I2C_SCL, I2C_SDA

STATUS LEDS

DL1 Green, R12 2K, LED_G, TP3

DL2 Red, R13 3K, LED_R, TP4

BOOT MODE SELECTOR

P4 — JA_PIN_Header 2

R15 10K

USB C

H1

VBUS, CC1, Dp1, Dn1, SBU1, CC2, Dp2, Dn2, SBU2, GND, MNT 1, MNT 2, MNT 3, MNT 4

R1 5K1, R4 5K1

USB_D_P, USB_D_N

USB4085-GF-A
2073-USB4085-GF-ACT-ND

SERIAL CONNECTION

SERIAL — JA_PIN_Header 6

P6 — JA_PIN_Header 3

SERIAL_TX, SERIAL_RX, SWCLK, SWDIO

3V3 LDO REGULATOR

U4 CAT6219-330TDGT3CT-ND

TP8, TP10

C11 1uF 16V, C12 10nF, C21 2.2uF 10V, R6 0R

BUZZER FEEDBACK

Q1 BC817W 1727-4843-1-ND

R5 0R, R7 1K, R8 10K, R9 1K, TP1, P1

PWM_TIM2

SWITCHES

R17 10K, C10 100nF, R18 10K, R16 10K, P7 — JA_PIN_Header 2

RESET, nRST, SLEEP_WAKEUP, SLEEP, BUTTON_IN_2, S2, BUTTON_IN_1, S1

BUCK BOOST CONVERTER

BUCKBOOST — TPS63070RNMR

VIN, VSEL, EN, PS/SYNC, L1, L2, VOUT, PG, FB, FB2, VAUX, PGND, GND

C30 10uF 16V, C31 10uF 16V, C25 10uF 16V, C29 10uF 16V, C22 10uF 6.3V, C23 22uF 10V, C32 22uF 10V, C33 22uF 10V, C26 10uF 6.3V, C27 100nF, C28 100nF

R22 10K, R19 100K, R20 680K, R21 130K

L1 max. Idc 0.3R

STLINK PROGRAMMER PORT

J3 — SFMC-107-L1-S-D

SWDIO, SWCLK, nRST

POWER PINS

POWER — Header 3X2 Vert

EEPROM

J2 — 25LC256T-E/SN

VCC, SO, CS, SCK, WP, SI, HOLD, VSS

R24 0R, R2 10K, R3 10K

SPI_CS_EEPROM, SPI_CLK_EEPROM, SPI_MOSI_EEPROM, SPI_MISO_EEPROM

RFID TRANSPONDER BASE STATION

IC1 — EM4095

VDD, SHD, FCAP, MOD, DMOD_IN, CDEC_OUT, VSS, DVDD, RDY/CLK, ANT1, ANT2, AGND, DMOD_OUT, CDEC_IN, DC2, DVSS

C7 100nF, C8 100nF, C9 10nF, C13 6.8nF, C14 100nF

ANTENNA — JA_PIN_Header 4

ANT_SHD_5V, ANT_DEMOD_5V

5V TO 3.3V LEVEL SHIFTER

B2 — NVT2002DP_118

GND, VREFA, A1, A2, EN, VREFB, B1, B2

ANT_DEMOD_3V3, ANT_SHD_3V3, ANT_LVLSHIFT_EN, ANT_DEMOD_5V, ANT_SHD_5V

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